La Scuola d'Italia / Dr. Huson / IB Math 12
9 November 2026

Nickname:

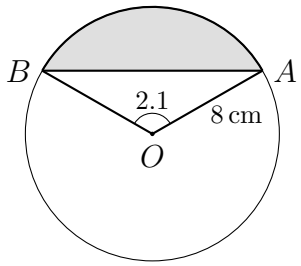
3.4 Pre-Test: Trigonometry Review

A calculator is allowed. Give answers correct to three significant figures, unless they are exact. Angles are in radians unless they are given in degrees. Diagrams are not to scale.

arc length $l = r\theta$	area of a sector $A = \frac{1}{2}r^2\theta$	
$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$	$c^2 = a^2 + b^2 - 2ab\cos C$	area $= \frac{1}{2}ab\sin C$

1. [Maximum mark: 11]

The diagram shows a circle with centre O and radius 8 cm. Points A and B lie on the circle, and $A\hat{O}B = 2.1$ radians.
The shaded region is bounded by the chord $[AB]$ and the minor arc AB.



(a) Find the length of the minor arc AB. [2]

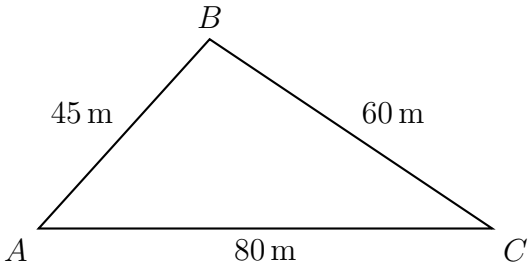
(b) Find AB. [3]

(c) Find the area of the shaded region. [4]

(d) Find the perimeter of the shaded region. [2]

2. [Maximum mark: 12]

A triangular garden ABC has sides $AB = 45\,\text{m}$, $BC = 60\,\text{m}$ and $AC = 80\,\text{m}$.



(a) Find the size of the largest angle of the garden. [3]

(b) Find the area of the garden. [2]

In a different triangle PQR, $PQ = 14\,\text{cm}$, $QR = 10\,\text{cm}$ and $Q\hat{P}R = 38°$.

(c) Find the two possible values of $P\hat{R}Q$. [3]

(d) Find the difference between the two possible lengths of PR. [4]

Nickname:

3. [Maximum mark: 9]

   (a) Write down the exact value of

      (i) $\sin \frac{4\pi}{3}$; [1]

      (ii) $\tan \frac{2\pi}{3}$. [1]

   (b) Given that $\sin \theta = \frac{2}{3}$ and $\frac{\pi}{2} < \theta < \pi$, find the exact value of

      (i) $\cos \theta$; [3]

      (ii) $\tan \theta$. [1]

   (c) Solve the equation $2 \cos x + \sqrt{3} = 0$ for $0 \leq x \leq 2\pi$. Give your answers as exact multiples of π. [3]

4. [Maximum mark: 13]

A Ferris wheel turns at a constant speed. The height, h metres, of a seat above the ground, t minutes after the seat passes its lowest point, is modelled by

$$h(t) = d - a\cos(bt), \qquad t \geq 0,$$

where $a,\ b,\ d > 0$. The lowest point of the seat is 2 metres above the ground and the highest point is 30 metres above the ground. One complete turn of the wheel takes 12 minutes.

(a) Find the value of a and the value of d. [2]

(b) Find the value of b. [2]

(c) Find the height of the seat 2.5 minutes after it passes its lowest point. [2]

(d) Find the length of time during one turn for which the seat is more than 25 metres above the ground. [3]

(e) Describe fully a sequence of transformations that maps the graph of $y = \cos t$ onto the graph of $y = h(t)$. [4]